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One-Shot Learning for Enhanced Cancer Type Detection: A Combined Approach of Gene Expression and Mutational Analysis

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Abstract

Cancer is a complex disease influenced by numerous concurrent genetic factors that result in diverse *tumor microenvironments* (TMEs) across different cancer types. Large-scale genomic projects, such as *The Cancer Genome Atlas* (TCGA), have underscored the need for molecular classification of cancer to enable more precise therapeutic strategies. Traditional machine learning (ML) approaches, while effective, predominantly rely on gene expression data and often overlook critical genomic alterations like Copy Number Alterations, Single Nucleotide Polymorphisms, and other mutational profiles, limiting the scope of biomarker discovery. In this study, we propose a novel ML-based method for cancer type detection that integrates both gene expression and genetic mutations to comprehensively characterize TMEs. By employing *Siamese Neural Networks* (SNNs), our method formulates the detection task as a similarity-based classification problem, enabling the model to generalize to new cancer types not encountered during training. This flexibility is particularly valuable in cancer genomics, where data scarcity and the need for continuous model updates are significant challenges. To further enhance *interpretability*, we introduce a robust explainability technique founded on *SHapley Additive exPlanations* (SHAP) values. This model-agnostic approach provides clear insights into the contributions of gene expression and mutational data, enabling a deeper understanding of the key factors driving cancer detection decisions.
Our experimental results demonstrate that integrating mutational profiles with gene expression data allows for more accurate cancer type detection and reveals biologically significant mutation patterns. These findings suggest that the proposed method has the potential to not only improve diagnostic accuracy but also identify key biomarkers relevant for immunotherapy success, addressing key limitations of existing approaches.

Key words: Deep learning; One-shot learning; Genomics; Cancer type prediction; Mutational profiles; Cancer gene markers

Introduction

Challenges of Machine Learning in Cancer Genomics. Cancer is a complex disease characterized by abnormal cell growth, which poses substantial challenges for scientific research, since both its insurgence and development are determined by many concurrent factors [1]. Its extreme complexity and heterogeneity is manifested in the unique genetic makeup or *molecular signature* of each individual tumor [2] which can vary with its location, stage, and ultimately patients. To better understand the genetic markers and molecular mechanisms underlying various types of cancer, large-scale genomic studies like *The Cancer Genome Atlas* (TCGA) [3, 4] have been conducted. These studies have analyzed and explored the genetic and omics profiles of extensive cohorts of patients affected by some of the most prevalent forms of cancer. The findings from these studies have increasingly shown that, while molecular profiles can accurately predict known cancer types, the full spectrum of cancer extends beyond traditional tumor classifications. This result highlights the importance of molecular classification, often based on gene expression analysis, for individual tumors, paving the way

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for precision cancer therapy that ensures specialized diagnoses and treatments tailored to each patient's unique molecular profile.

From a *genomic* perspective, gene expression has been largely studied for multiple purposes, including identifying driver cancer genes, forecasting patient prognosis and therapy success, and biomarkers identification [5, 6]. Statistical testing methods involve the use of hypothesis tests to determine if there are statistically significant differences or associations within the data. Common examples include *t-tests*, *chi-squared* tests, and *ANOVA* (Analysis of Variance), which are used to compare gene expression levels across different conditions or patient groups. These methods are grounded in well-established statistical theories and provide interpretable results, making them a valuable tool in genomics research. However, they have limitations, particularly when dealing with high-dimensional data such as gene expression profiles, where multiple testing corrections are necessary to control the false discovery rate, potentially reducing statistical power and missing important associations [7, 8, 9, 10]. Conversely, ML methods are designed to handle large and complex datasets. They can model non-linear relationships and interactions between genes, making them particularly suitable for identifying patterns in gene expression data that might be missed by traditional statistical approaches. Such methods have been applied successfully to classify cancer types, predict patient outcomes, and identify potential therapeutic targets by learning from large-scale genomic datasets [11, 12, 13, 14, 15, 16].

While ML methods have significantly advanced the identification of cancer biomarkers, they still face several critical limitations. First, they tend to focus predominantly on gene expression data [17], often overlooking the valuable insights provided by *gene mutations*, such as *Copy Number Alterations* (CNAs), *deletions*, *insertions*, and *Single Nucleotide Polymorphisms* (SNPs). This narrow focus on gene expression limits the scope of biomarker discovery, as it fails to capture the full complexity of genomic alterations that drive cancer progression. Gene mutations play a critical role in oncogenesis by altering gene function and regulation, contributing to cancer heterogeneity, and impacting patient prognosis and response to therapy. Incorporating these genomic variations into ML models can provide a more comprehensive understanding of cancer biology and lead to the identification of more robust biomarkers. Recent studies have demonstrated that integrating gene mutation data with gene expression analysis can significantly improve the accuracy of cancer classification, prognosis prediction, and therapeutic target identification. For instance, large-scale projects like the *Pan-Cancer Analysis of Whole Genomes* (PCAWG) have highlighted the critical role of mutations in both coding and non-coding regions, revealing that many driver mutations occur in non-coding regions that regulate gene expression [18]. Additionally, CNAs have been shown to be a key factor in cancer development, influencing gene dosage and disrupting normal cellular functions, particularly in breast cancer and other common malignancies [19]. Furthermore, SNPs and other small-scale mutations have been linked to the disruption of gene regulatory elements, which can lead to the dysregulation of oncogenes and tumor suppressor genes, further underscoring the need to integrate these data into ML models [20, 21]. Several analysis of germline mutations have revealed significant associations between specific mutations and cancer risk, highlighting the importance of these variations in understanding cancer etiology and progression [22]. Moreover, advances in whole-genome sequencing have uncovered the complex interplay between somatic and germline mutations in driving cancer, providing new avenues for targeted therapies [23]. It is therefore evident that, as the field of cancer genomics continues to evolve, it is crucial for ML approaches to incorporate these diverse forms of genomic data to enhance biomarker discovery and ultimately improve patient outcomes.

Another major limitation of current ML and Deep Learning (DL) approaches in the context of cancer research is their inherent "black-box" nature. These models often lack transparency, making it difficult to understand the underlying reasons for their performance on specific datasets. This lack of "explainability" is a significant drawback when dealing with biological data, where the elucidation of underlying mechanisms is not only desirable but essential. In critical domains like cancer analysis, where decisions can directly impact patient outcomes, the inability to explain why a model works, or how it reaches its conclusions, poses a substantial risk. It can undermine trust in the model's predictions and complicate the validation of results within the broader scientific community. Furthermore, the nature of supervised learning models as classifiers introduces additional challenges in biological contexts. These models are generally designed to solve classification problems, requiring a predefined set of classes that the model is trained to recognize. However, in rapidly evolving fields such as cancer genomics this approach proves to be inflexible. Whenever a new class needs to be incorporated, the entire model must be retrained with updated biological data. This process is not only computationally expensive but also introduces delays in the application of these models in clinical settings, where timely and accurate analysis is crucial. Such type of approach often overlooks the potentially more informative aspects of biological data, such as the discovery of novel patterns or the understanding of underlying biological mechanisms. This limitation underscores the need for more advanced methodologies that can integrate diverse types of genomic data and adapt more dynamically to the evolving landscape of cancer research.

Role of Genetic Mutations in Deciphering Tumor Microenvironment. A comprehensive study of cancer and the identification of reliable biomarkers require a thorough understanding of the *tumor microenvironment* (TME), which represents the intricate cellular environment in which a tumor develops, encompassing a variety of cells such as immune cells, stromal cells, and blood vessels. It plays a pivotal role in shaping tumor behavior, influencing not only its growth and progression but also its response to therapies, so a precise understanding of if is essential for developing targeted treatments and improving patient outcomes. Closely tied to the TME is the concept of *tumor mutational burden* (TMB), which refers to the total number of genetic mutations, such as CNAs, deletions, insertions, and SNPs, present within the cancer cells' DNA, which contribute to the complexity of the cancer genome. The relationship between the TME and TMB is intricate and multifaceted [24, 25, 26, 27, 28]. For instance, the immune cells within the TME, such as T cells, can recognize tumor cells through neoantigens produced by somatic mutations. A higher TMB often results in an increased production of these neoantigens, potentially enhancing the visibility of the tumor to the immune system. Tumors with a higher TMB, enriched with mutations like CNAs and SNPs, may be more likely to elicit a robust immune response, making them more responsive to immunotherapies like immune checkpoint inhibitors. Moreover, the TME itself can influence the mutational landscape of the tumor. Inflammatory conditions within the TME can promote DNA damage, leading to the accumulation of mutations. Additionally, stromal and immune cells within the TME can modulate DNA repair processes in tumor cells, affecting their overall mutational burden. Conversely, certain components of the TME, such as immunosuppressive cells, can create an environment that supports tumor growth and mediates resistance to therapy, even in tumors with a high TMB.

This underscores the importance of understanding the interplay between the TME and TMB for advancing cancer research and developing more effective treatments. Integrative analyses that combine genetic mutation profiles with TME characteristics can yield novel biological insights and identify features relevant to tumor classification and TME functions. For example, specific mutation frequency patterns across different types of mutations can serve as valuable signatures for cancer classification or biomarker discovery, offering insights into the prognostic or predictive value of these mutations. Furthermore, investigating the functional implications of these mutations within the TME context can elucidate their roles in key biological pathways, such as immune modulation

and angiogenesis, which are crucial for tumorigenesis and therapeutic resistance. By prioritizing genes involved in these pathways, researchers can guide the development of targeted therapies that disrupt the supportive environment facilitating tumor growth.

Our contribution. In this work, we present a novel machine learning-based approach for cancer type detection, which addresses key limitations in current cancer genomics methods. A central innovation of our approach is the integration of gene expression data with genetic mutations (CNAs, insertions, deletions, and SNVs), to capture a more comprehensive understanding of the TME [29, 30]. This combined data approach allows the proposed ML model to "learn" the differences in TME expression profiles across various cancer types, thereby enhancing the detection and classification of tumors. This approach is motivated by the growing importance of immunotherapy, a treatment strategy that relies on the immune system's ability to recognize and destroy cancer cells. A key challenge in immunotherapy is overcoming the suppressive effects of the TME, which often diminishes the immune response against tumors [31]. Currently, the TMB, defined as the average number of mutations per 1k bases, is the only approved indicator in the U.S. for predicting immunotherapy success [32]. However, TMB has shown inconsistencies and lacks statistical significance in some cases. To improve upon this, we incorporate a new indicator known as *persistent TMB* (pTMB), which focuses on mutations that are either heterozygous or present in multiple copies [33]. These mutations are less likely to be lost during tumor evolution, making them more reliable for predicting therapeutic outcomes. Building on this integrated data approach, we reformulate the cancer detection problem using a *similarity-based classification* paradigm [34]. Specifically, we employ *Siamese Neural Networks* (SNNs) [35] to learn a similarity function from pairs of cancer samples. Unlike traditional classification models that assign labels based on predefined classes, our model learns to evaluate the similarity between samples, allowing it to generalize to new cancer types not encountered during training. This flexibility is particularly advantageous in cancer detection, where *(i)* the diversity of cancer types and the scarcity of labeled data often pose significant challenges, and *(ii)* the addition of a new cancer class would not require retraining the entire model, provided that the dataset used for the training is truly significant.

To further enhance the interpretability of the proposed method, we introduce an innovative explainability approach by integrating *SHapley Additive exPlanations* (SHAP) values into our SSNs framework [36, 37]. SHAP values are a robust explainability technique that is both model-agnostic and theoretically grounded, providing clear and interpretable insights into model predictions. This explainability is crucial in cancer type detection, where understanding the model's decision-making process can reveal which genes and mutational patterns are most relevant for specific cancer types.

Although SSNs have been applied in various fields, their use in bioinformatics and cancer genomics remains relatively recent and limited [35, 38, 39]. The approach proposed in this paper represents a significant evolution of [17], based on SNNs but focusing solely on gene expression as the feature set, which, while informative, does not capture the full complexity of the cancer landscape. Our approach advances these efforts by integrating both gene expression and genetic mutation data, including CNAs, structural variations, and other mutational information, into a unified framework. Results obtained from experiments conducted to assess the effectiveness of the proposed approach, show that this integration allows for a more complete characterization of the TME, enabling more accurate and robust cancer type detection, and also provides valuable insights into the underlying biological mechanisms, potentially identifying stable indicators for the success of immunotherapy.

In summary, the main contributions of this work are as follows:

- A novel ML-based method for cancer type detection that leverages the strengths of SNNs for discovering a set of type-agnostic marker genes that, combined with their corresponding mutational profiles, can be used for for establishing the similarity or dissimilarity between samples from distinct TMEs of various cancer types; this method integrates information from both gene expression and genetic mutations to identify mutation patterns specific to each cancer type, many of which have been validated by existing literature; tese findings pave the way for establishing unique *mutation signatures* for different cancer types.
- A novel explainability method for SNNs, founded on the theory and the functioning of SHAP values, which is model-agnostic, theoretically grounded, and offers clear, interpretable insights into model predictions, making it a powerful tool for understanding the underlying mechanisms in cancer detection.

Related works

This section provides a comprehensive discussion of the most relevant studies in the literature on ML-based cancer detection. In doing so, we will also highlight the key technical differences between these studies and the method proposed in this paper, emphasizing the advancements and innovations introduced by our approach.

Earlier studies in cancer detection focused primarily on distinguishing between cancerous and normal tissue samples using neural network. In [40], for example, the authors proposed a Deep Learning (DL) model that exploited gene expression data to identify whether a tissue sample was cancerous or not, achieving high accuracy. However, this model was not designed to classify specific cancer types, limiting its applicability in more nuanced cancer detection tasks. In response to the growing need for models that can differentiate among specific cancer types, several recent studies have applied DL models to classify tumor samples. For example, [41, 42] converted *1-dimensional* gene expression data from *TCGA* into *2-dimensional* image-like inputs to leverage the power of convolutional neural networks (CNNs) for classifying 33 different cancer types. This approach leveraged the image-processing strengths of CNNs, allowing the model to capture spatial patterns in the data that might be informative for the classification. Building on this concept, methods like [43, 44, 45] incorporated gene correlations into the 2-dimensional images to preserve the inherent relationships between genes when transforming the data into image format, so improving the model's ability to classify cancer types accurately. Despite these advancements, these approaches still primarily relied on large, well-annotated datasets such as *TCGA*, which limited their ability to generalize to new cancer types, particularly those not included in the training data. To overcome such limitations, in [46] the authors introduced a 1D-CNN model directly applied to 1-D gene expression data to classify 33 *TCGA* cancer types and an additional normal tissue type. This approach proved that a 1D-CNN could achieve performance comparable to 2D-CNNs but with significantly fewer parameters, reducing the risk of overfitting and improving the model's robustness in generalizing to new data.

Although these strategies have demonstrated impressive performance, they also present several limitations that limit their broader application in personalized cancer classification, particularly within the framework of precision oncology [47]. First, their high classification accuracy is heavily dependent on the availability of large-scale, well-annotated tumor datasets. Furthermore, for rare tumors, collecting enough samples to train a model with satisfactory performance is often unfeasible. Finally, as cancer classification moves towards more refined, molecular-based characterizations, there is a growing need for models that align better with the nature of precision oncology, focusing on the nuanced molecular characteristics of tumors rather than broad classifications. Traditional classification approaches may fall short in this context since they often oversimplify the complexity and heterogeneity of tumors.

A notable attempt to overcome these limitations is presented in [48], where a k-nearest neighbors-based graph attention *few-*

shot learning [49] model, named *AffinityNet* and designed to address data scarcity in cancer type prediction, was proposed. However, *AffinityNet* represented an advance in the field but was not a *one-shot learning* [50] model, i.e., trained to recognize and classify objects based on only a single example (or very few examples) of each category, with many advantageous in scenarios where collecting large datasets is impractical, such as in rare diseases or when obtaining labeled data is costly. Furthermore, its scope was also limited since it was trained on only two primary cancer types and so unable to predict cancer types not included in the training data. Moreover, it did not provide an interpretability mechanism, leaving the underlying biological markers and functions unexplored.

Starting from the awareness of such limitations, in [17] *CancerSiamese* was introduced, a model employing Siamese Convolutional Neural Networks (SCNNs) to address the challenge of predicting new cancer types unseen during training. The authors proposed a one-shot learning strategy that required only a single "support" sample from each new cancer type, significantly reducing the need for large datasets per cancer type. *CancerSiamese* classified query samples by comparing them to a set of support samples, one for each cancer type, thus leveraging a powerful similarity metric learning model. The success of this system reinforced the potential of SCNNs already proved in other bioinformatics and medical applications, including drug response similarity prediction [51], sequence embedding and alignment [52], and cell type identification [53].

However, despite its innovative approach, *CancerSiamese* mainly focused on the efficiency of the model as a diagnostic tool. First, the model solely relied on gene expression data, without taking advantage of the mutational information available. Secondly, the explainability method chosen to investigate the model's behavior, i.e., the *Guided Backpropagation Saliency Maps*, only provided a global explanation, identifying which gene expression were useful on a general basis, but failing in identifying features that were import to specific cancer types. In this work, we extend the *CancerSiamese* approach by proposing a novel model that leverages the strengths of SNNs to incorporate both gene expression and genetic mutations. We assume that a set of type-agnostic marker genes, combined with corresponding mutational profiles, can define the similarity or dissimilarity between samples from the same or different TMEs of cancer types. As future perspective, this approach could allow our model to generalize across cancer types while significantly reducing the need for large datasets per type. By focusing on a similarity-based classification paradigm, our model can, on one hand, predict new cancer types unseen during training, making it a powerful tool for advancing cancer detection and treatment strategies in the era of precision oncology. On the other hand, it enables the identification of mutation patterns specific to particular cancer types, which are consistent with findings in the current literature. As we will see, we achieved better results with those of *CancerSiamese* by adding more information and for a larger number of classes. Specifically, we achieved an accuracy of 85.3% using a 24-way with the addition of mutations, while *CancerSiamese*, with the 10-way (the highest N-way they used), achieved an accuracy of 84.59% with only 19 tumors and only gene expression.

Table 1 provides a comparative overview of discussed approaches to cancer detection, focusing on key technical aspects such as the *usage* of gene expression and genetic mutation data, the *classification paradigm* employed, the inclusion of *explainability* mechanisms, and the *range* of cancer types considered. As we can see, our proposed approach stands out by integrating both gene expression and genetic mutation data within a similarity-based classification paradigm using SNNs. Additionally, the use of SHAP values for explainability allows for a clear interpretation of the model's predictions, providing insights into which features are most influential in determining similarity or dissimilarity between samples.

Background

Here, we briefly recall the needed background to understand the article. We assume that the reader is familiar with ML and Genomics [54]. Specifically, we will first provide a detailed overview of SNNs, explaining the advantages they offer to ML tasks. Following this, the section will delve into the application of SHAP values as a powerful tool for enhancing the explainability of ML models.

Siamese Neural Networks

Siamese Neural Networks (SNNs) are a specialized type of neural network (NN) architecture designed for determining the similarity or relationship between two comparable inputs. Unlike traditional NNs that produce a classification label as output, SNNs generate *embeddings*, or vector representations, for each input, which are then be compared using a *similarity function*, typically a *distance metric* such as the Euclidean distance, to determine the degree of similarity between the two inputs. The architecture of SNNs typically consists of two identical sub-networks that share the same weights and parameters. These sub-networks process the two input data samples in parallel and produce their respective embeddings. The core idea is that similar input samples will yield embeddings that are close to each other in the vector space, while dissimilar inputs will produce embeddings that are more distant from one another.

Such models have proven particularly effective in scenarios where labeled data is limited, which is common in fields like bioinformatics, where acquiring large, labeled datasets can be challenging. One key advantage of SNNs is their ability to learn from few examples, making them well-suited for tasks like *one-shot learning*, where the model must correctly identify or categorize a new input based on only one or very few training examples [50, 35].

One-shot learning is a subset of a broader class of learning paradigms known as *N-way shot learning*, where "N" refers to the number of examples provided per class during the training. In one-shot learning, N is set to one, i.e., the model is trained to recognize and classify objects or data points based on just a single example per class. This approach is particularly advantageous in situations where data is limited, such as in rare disease classification, where obtaining numerous labeled examples for training is challenging. *N-way shot learning* encompasses not just one-shot learning but also few-shot learning and other variations where N can be any small number greater than one. The central idea behind these methods is to enable the model to generalize effectively from a minimal amount of training data, making them extremely valuable in domains where data collection is expensive or difficult.

SNNs are particularly well-suited for N-way shot learning due to their architecture, which focuses on learning the similarity between pairs of inputs rather than relying on large datasets to perform classification. In the context of N-way shot learning, SNNs are used to compare an unseen test sample against N samples (one from each class) provided during training. For instance, in a one-shot learning scenario, an SNN might compare a new sample with a single prototype from each class, determining which class the sample is most similar to based on the learned embeddings. As the value of N increases, the SNN is still able to effectively generalize by leveraging the similarity between the new sample and multiple samples from each class. This ability to generalize from minimal data is what makes SNNs and other similar models particularly powerful in few-shot and N-shot learning scenarios [35, 55].

The training of SNNs involves optimizing a specific loss function that guides the learning process. The most commonly used loss function in SNNs is the *contrastive loss*, which directly measures the similarity or dissimilarity between two inputs, and defined as:

$$L(y, d) = (1 - y)\frac{1}{2}(d)^2 + y\frac{1}{2}(\max(0, m - d))^2$$

Table 1. Comparison of the ML-based systems for cancer detection.

Model	Gene Expression	Genetic Mutations	Paradigm	Cancer Types Considered
[40]	Yes	✗	Classification	Normal vs. Cancer
[41, 42]	Yes (2D-converted)	✗	Classification	33 Cancer Types
[43, 44, 45]	Yes (2D-converted)	✗	Classification	33 Cancer Types
[46]	Yes	✗	Classification	33 Cancer Types + Normal
[48]	No	✗	kNN-based Few-shot	2 Cancer Types
[17]	Yes	✗	One-shot Learning (SCNN)	19
Our Approach	Yes	✓	One-shot Learning (SCNN)	24

where y is the binary label which indicates whether the input pair of samples is similar (0) or dissimilar (1), d is the distance between the two embeddings, m is the *margin*, which is a hyperparameter that sets the minimum distance between dissimilar pairs.

Another popular loss function is the *triplet loss*, particularly effective in ranking-based tasks, and works by comparing an *anchor input* with a *positive sample* (same class) and a *negative sample* (different class). It is designed to ensure that the distance between the anchor and the positive sample is smaller than the distance between the anchor and the negative sample by a specified margin:

$$L(a, p, n) = \max(d(a, p) - d(a, n) + m, 0)$$

where a, p, and n are the anchor, positive, and negative samples, respectively; d represents the distance function (e.g., the Euclidean distance), and m is the margin that defines how much closer the positive sample should be compared to the negative sample.

These loss functions are crucial for training SNNs to effectively distinguish between similar and dissimilar input samples, which is the core of their utility in one-shot learning and related tasks.

SHAP Values for Feature Importance Calculation

In the field of ML *interpretability*, *SHapley Additive exPlanations* (SHAP) values [56, 37] have emerged as a prominent method due to their strong theoretical foundation and practical versatility. SHAP values provide a robust framework for attributing the contributions of individual features to model predictions, and derive from the Shapley values concept in game theory, which ensures fair distribution of payouts based on each player's contribution to a collective outcome. This theoretical grounding provides a solid basis for understanding the role of each feature in a model's prediction. Specifically, the Shapley value for a specific feature is computed as the average marginal contribution of that feature across all possible subsets of features. Formally, this is mathematically defined as:

$$\phi_i(x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (1)$$

where x is the analyzed instance, N represents the set of all features of x, S is a subset of N excluding the feature i and $f(S)$ denotes the model's prediction considering only the features in subset S.

One of the most significant advantage of SHAP values is their model-agnostic nature. This versatility allows SHAP values to be applied across various types of ML models, and enables meaningful comparison of feature importance across different model architectures, providing a unified approach for model interpretability. They also possess several desirable properties that enhance their utility:

- *efficiency*: it guarantees that the sum of SHAP values for all features equals the difference between the model's prediction for an instance and the average prediction over all instances:

$$\sum_{i=1}^{|} N|\phi_i(x) = f(x) - E[f(x)] \quad (2)$$

where $|N|$ is the number of features, $f(x)$ is the model's prediction for x, and $E[f(x)]$ is the expected value of the model's prediction over all instances; this ensures that the full impact of the prediction is accurately accounted for by the feature contributions.
- *symmetry*: if two features contribute equally to the prediction, they receive the same SHAP value; this ensures equitable attribution of contributions among features with equivalent impacts.
- *additivity*: it allows for the summation of SHAP values to understand the total contribution of features; this property enables coherent aggregation and comparison of feature effects:

$$f(x) = E[f(x)] + \sum_{i=1}^{|} N|\phi_i(x). \quad (3)$$

- *null feature*: features that do not influence the prediction are assigned a SHAP value of zero; this ensures that features with no effect are accurately represented in the attribution process.
- *consistency*: if a model's changes lead to an increased contribution from a feature (while other factors remain constant), the SHAP value for that feature should not decrease, ensuring logical consistency in feature importance evaluation.

Methods

We remark that the main goal of this study is to overcome the limitations of current cancer type detection systems, as discussed in the *Introduction*, and to develop a novel ML-based method that utilizes the strengths of SNNs to effectively learn and model the TME through the combined use of gene expression data and genetic mutations. The research questions driving our investigation are:

Key Research questions

1. How can we design a method that leverages cancer type-agnostic marker genes to study TMEs and discover mutation patterns unique to each cancer type, thus enhancing cancer type detection?

2. How can we create a robust model that provides clear, interpretable insights for each cancer type, ensuring that the key genetic factors driving predictions are understood and effectively applied in clinical practice?

This section details the methodology followed to answer these questions. First, we provide details about the architecture of the proposed SNN, the specific one-shot learning approach adopted, and the experiments carried out to assess the effectiveness of the proposed method. Finally, we describe and discuss the results obtained, highlighting the advantages of our approach in order to suggest potential directions for future research.

The proposed SNN

In the following, we will denote with S the proposed SNN. As discussed in Section *Background*, a SNN consists of two identical subnetworks. In our case we can denote the structure of S as follows:

$$\mathtt{S} = [\mathtt{S_q}, \mathtt{S_s}]$$

where S_q (resp. S_s) is *query* (resp. *support*) subnetwork of S.

The proposed S is defined to take as input the feature vectors $fv(q)$ and $fv(s)$ of two patient samples q (*query* patient simple) and s (*support* patient simple) and calculate their similarity $\mathtt{S}(fv(q), fv(s))$. However, the estimation of this similarity is obtained in several intermediate steps. First, given the patient feature vectors $\mathtt{fv}(q)$ and $\mathtt{fv}(s)$, the query subnetwork S_q computes the *embedding* $\mathtt{S_q}(\mathtt{fv}(q))$ of q, and the support subnetwork S_s computes the *embedding* $\mathtt{S_s}(\mathtt{fv}(s))$ of s. Then, the *L2 distance* between the embeddings $\mathtt{S_q}(\mathtt{fv}(q))$ and $\mathtt{S_s}(\mathtt{fv}(s))$ is computed as follows:

$$\mathrm{L2}\Big(\mathtt{S_q}(\mathtt{fv}(q)), \mathtt{S_s}(\mathtt{fv}(s))\Big)$$

Such a distance, also called *Euclidean distance*, is used as a measure of similarity between two points in a multidimensional space, and is calculated as the square root of the sum of the squared differences between corresponding coordinates. Formally, given two points q and s, the distance of q and s is calculated as follows:

$$d(q, s) = \sqrt{\sum_{i=1}^{n} (q_i - s_i)^2}$$

The intrinsic meaning is that the smaller the Euclidean distance between $fv(q)$ and $fv(s)$, the greater the similarity between q and s.

In the following, we provide details about the architecture and training process of our proposed SNN (S). Notice that the architecture of S is an extended version of the one proposed in [17], which has been modified multiple times (in terms of hyperparameters, type, and number of layers) throughout various experiments to optimize performance. Figure 1 illustrates the architecture that achieved the best results. Below, we detail the structure of the two subnetworks of S. As shown in Figure 1, S consists of two identical CNNs applied to the query and support patient samples individually, followed by a similarity metric network. We adopted the 1D-CNN proposed in [17] due to its simplicity and high accuracy for TCGA cancer type prediction, as also proved in [17]. However, to effectively capture the complex patterns within the genetic mutations, empirical observations from our several preliminary experiments proved the need for a "deeper" version of the SNN proposes in [17], proved inadequate for learning the intricate patterns of genetic mutations, as evidenced by our experimental results. After numerous attempts, we discovered that deepening the network was essential for effectively capturing both gene expression and mutation information, thereby enabling a more accurate TME modeling.

Specifically, following the first 1D convolution layer, we added four consecutive 1D convolution and max-pooling layers, followed by a flatten layer. The ReLU activation function was applied to the first four 1D convolution layers, while a sigmoid activation function was used for the final 1D convolution layer. For the similarity metric network, an element-wise *L2 distance* (using a contrastive loss function) was computed between the two feature vectors generated by the 1D-CNNs, with the output passed through two consecutive fully connected layers followed by a sigmoid node to determine whether the input pairs belong to the same cancer type.

As discussed in Section *Background*, one of the key advantages of using SNNs is their ability to leverage the *One-Shot Learning* strategy, which has been proven effective in identifying new classes based on just one or a few examples. The concept is to extract rules from previously observed classes, learning patterns and similarities, rather than fitting the ML model to fixed classes, thus enabling the classification of previously unseen classes using a single instance.

Here, we provide details about the implementation of S based on one-shot learning and the methodology used to evaluate performance on new cancer type classes without retraining it.

The *training* is based on a *pair-based* approach on gene expression and genetic mutations data to differentiate between various cancer types. During the training phase, the model processes pairs of samples, each consisting of an *query* and a *support* sample. These samples are selected from the dataset and can either belong to the same cancer type (*positive pairs*) or different cancer types (*negative pairs*). As explained above, for each pair, the query and the support subnetwork of S map the query and the support samples into a lower-dimensional embedding space designed to capture the essential features of both the gene expression profiles and the genetic mutations patterns that are relevant for distinguishing between different cancer types. Once the embeddings for both the query and support samples are generated, the model computes the Euclidean distance (*L2 distance*) between these two embeddings. This distance serves as a measure of similarity between the two samples: a smaller distance indicates higher similarity, i.e., both samples are likely to be affected by the same cancer type, while a larger distance indicates lower similarity, i.e., the samples likely to be affected by different cancer types. The goal of the training is to minimize a *contrastive loss* function, which forces the network to produce small distances for positive pairs and large distances for negative pairs. Indeed, the contrastive loss is defined in such a way that it penalizes the model when the distance between embeddings of positive pairs is too large or when the distance between embeddings of negative pairs is too small. This drives the model to learn a feature space where similar samples (same cancer type) are close together, and dissimilar samples (different cancer types) are far apart.

During the *performance evaluation* phase, S is used in a one-shot learning scenario. Specifically, the network S receives a new, previously unseen cancer type, represented by a single support sample. The goal is to classify a query sample by comparing its embedding to the embedding of the support sample. To determine the cancer type of which a *query* patient sample q is affected, the proposed SNN S exploits a reference set of *support* patient samples $\mathcal{S} = \{s_1, \ldots, s_m\}$. For each pair of patient feature vectors $\mathtt{fv}(q)$ and $\mathtt{fv}(s_i)$ with $s_i \in \mathcal{S}$, S first computes the *embeddings* $\mathtt{S_q}(\mathtt{fv}(q))$ and $\mathtt{S_s}(\mathtt{fv}(s_i))$, and then the *L2 distance* as calculated as follows:

$$\mathrm{L2}\Big(\mathtt{S_q}(\mathtt{fv}(q)), \mathtt{S_s}(\mathtt{fv}(s_i))\Big)$$

Then, the cancer type $c(q)$ of q is assigned by looking at the cancer type of the support sample s_i in S that has the minimum L2 distance from q. We denote this situation with $q \sim s_i$. Formally, given a set of *support* samples $\mathcal{S}$ and a query sample q, the cancer type $c(q)$ of q is assigned by applying the following rule:

$$c(q) = c(s) \iff s = \arg\min_{s_i \in \mathcal{S}} \mathrm{L2}\Big(\mathtt{S_Q}(\mathtt{fv}(q)), \mathtt{S_S}(\mathtt{fv}(q))\Big) \iff q \sim s$$

The *performance evaluation* is summarized by the Algorithm 2.

This approach enables S to generalize to new cancer types with few examples, leveraging the feature space learned during training.

As we can see in Algorithm 1, the training phase of S begins by initializing the S's parameters using the `InitializeSiamese` function (line 1). The substantial size of S posed challenges when training it from scratch, leading to issues such as training instability and subpar convergence, which ultimately affected the robustness of predictions. To overcome these challenges, as also proposed in [17], we implemented a transfer learning strategy in the development of S. Initially, we trained a 1D-CNN on TCGA cancer datasets for

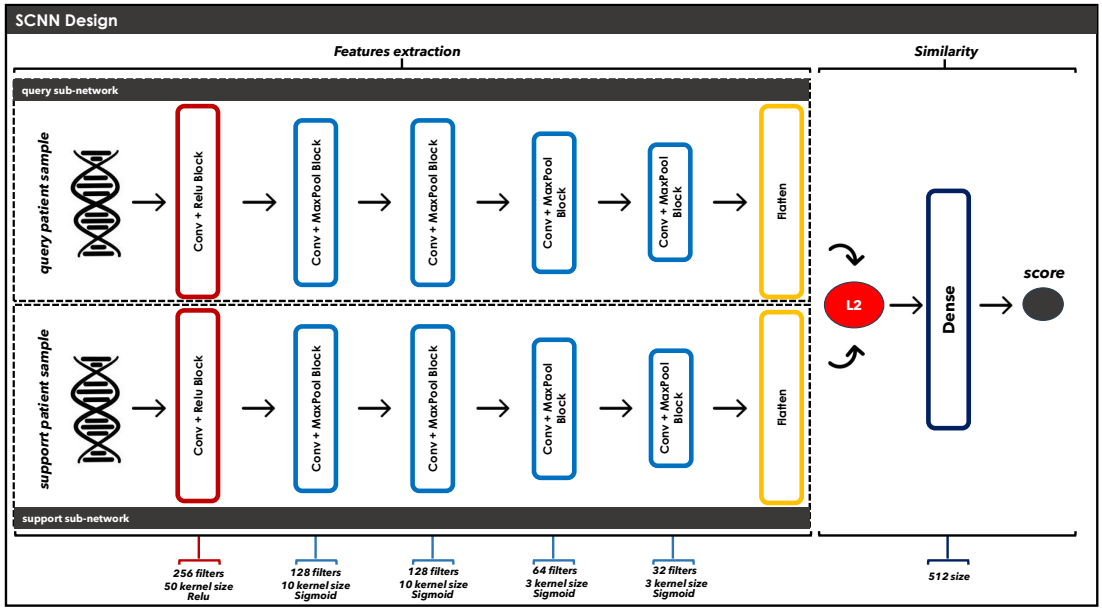


Figure 1. The proposed Siamese Convolutional Neural Network (SCNN) S.

Algorithm 1: Siamese NN S Training phase

Input : Training pairs (query, support, label), number of epochs num_epochs
Output : Trained Siamese Network S

```
1  S = [S_q, S_s] ← InitializeSiamese();
2  for epoch = 1 to num_epochs do
3      foreach (query, support, label) training_pairs do
4          query_embedding ← S_q(fv(query));
5          support_embedding ← S_s(fv(support));
6          distance ← calculate_L2_distance(query_embedding,
             support_embedding);
7          loss ← contrastive_loss(distance, label);
8          loss.backward();
9          optimizer.step();
```

Algorithm 2: Siamese NN S Evaluation with One-Shot Learning

Input : Trained S = [S_q, S_s], support set $\mathcal{S}$ containing n samples, query sample query
Output : Prediction result: Most Similar Cancer Type in $\mathcal{S}$

```
1  query_embedding ← S_q(query);
2  foreach sample in S do
3      sample_embedding ← S_s(sample);
4      distance[sample] ← calculate_L2_distance(query_embedding,
         sample_embedding);
5  most_similar_sample ← argmin(distance);
6  predicted_cancer_type ← cancer_type(most_similar_sample);
7  return predicted_cancer_type;
```

the purpose of classifying the 24 cancer types described in Table 2. After completing this training, we removed the classification layer and utilized the remaining pretrained 1D-CNN as the foundation for our S (see Figure 1). In the subsequent training of S, these 1D-CNN components were further refined to optimize the prediction of sample similarity. Since the 1D-CNN had already been trained to effectively capture the expression features critical for cancer classification, its weights were already close to those ideal for one-shot learning. Therefore, by initializing S with this pretrained network, we achieved greater stability during training and faster convergence. The process is then iterated over a specified number of epochs. For each epoch, the algorithm processes pairs of query and support samples along with their corresponding labels. First, the feature vector of the query sample is passed through the network to generate the `query_embedding` (line 1). Similarly, the feature vector of the support sample is processed to generate the `support_embedding` (line 5). The L2 distance between these two embeddings is then computed (line 4), which serves as the basis for calculating the contrastive loss (line 7), which is then back-propagated through S, and the S's parameters are updated accordingly.

SHAP-Enhanced SNNs: a novel Mathematical perspective

Let S be a SNN, and let S(x_i, x_j) be the a *similarity score* computed by S for a pair of input samples (x_i, x_j), where x_i (resp. x_j) represents the *feature vector* corresponding to the i-*th* (j-*th*) cancer sample. The score S$(x_i, x_j) \in [0, 1]$ reflects the degree to which x_i and x_j are likely to belong to the same cancer type. The goal of integrating SHAP values into this framework is to quantify the contribution of each feature f_k to S(x_i, x_j). Due to computational complexity, approximations by `DeepExplainer`[1] from the `SHAP` library are used.

We recall that our model S takes in input two samples, x and y, i.e. the pair of their *feature vectors*, $fv(x)$ and $fv(y)$. Given a feature i, this feature assumes two different values, one in sample x and one in sample y, which we call $fv_i(x)$ and $fv_i(y)$, respectively. However, in the calculation of the SHAP values, $fv_i(x)$ and $fv_i(y)$ are treated independently and they they bot get two different SHAP values, which we call $\phi_i(x)$ and $\phi_i(y)$. $\phi_i(x)$ and $\phi_i(y)$ are obtained following the classic definition of SHAP values addressed in Subsection . Nevertheless, our final goal is to obtain a unique SHAP value summarizing the *importance* of feature i for a given pair. We define the unified SHAP value of feature i for the pair $p = (x, y)$, as:

$$\phi_i^{p=(x,y)} = \frac{|\phi_i(x)| + |\phi_i(y)|}{2} \tag{4}$$

Notice that, in our context, where the input is a pair of which the SNN must calculate the similarity, one SHAP value $\phi_i^{p=(x,y)}$ measures the contribution of a feature i to the output, but taking into account of the "marginal" *contribution directions* with respect to each one of two samples, i.e., $\phi_i(x)$ and $\phi_i(y)$. By definition, positive SHAP values indicate that the feature increases the similarity

1 https://shap.readthedocs.io/en/latest/generated/shap.DeepExplainer.html

score, while negative values suggest it decreases the score. However, the absolute value of the SHAP score reflects the *magnitude* of the feature's contribution, regardless of direction. By computing the mean of the absolute SHAP values for each feature across both inputs in the pair, we capture the overall influence of a feature in the paired-input setting. While this method sacrifices the straightforward interpretation of the SHAP value as the marginal contribution to the model's prediction output, it reflects the "overall impact" of a feature on the model's prediction for both samples in the pair. For example, if feature f_1 has SHAP values of -0.5 for one sample and 0.5 for the other, summing the SHAP values would give 0, implying no net effect. However, in absolute terms, the feature is still contributing significantly to the output, just in opposite directions. By summing the absolute values, we capture the *overall influence* of the feature, regardless of whether it increases or decreases the similarity score, and suggesting how strongly the feature influences the output, without canceling out contributions of opposite signs.

Let us consider N pairs of samples. The definition of the SHAP value *importance* of a feature i in a given pair of a samples $p = (x, y)$ provided above, is used to define the *global importance* SHAP value of feature i (Φ_i) as the mean value of $\phi_i(p)$ across all pairs. Formally:

$$\Phi_i = \frac{\sum_{p=1}^{N} \phi_i^p}{N} \tag{5}$$

Our final goal was to identify the most important features for each cancer type, c. To this end, for each cancer type we built an *ad-hoc* set of pairs P_c to use in the calculation of the cancer-specific *global feature importance* $\Phi_i(c)$ of any feature i. The cancer-specific datasets were built as follows: for a given cancer type c, we *(i)* extracted all the samples (*patients*) relative to that cancer type, and *(ii)* generated two pairs for each selected sample, i.e. a *positive* one, by pairing it with another sample from the same cancer type, and a *negative* one, pairing it with a sample from a different cancer type. By considering all the available samples, we take advantage of all the available data. At the same time, by using each sample in both positive and negative samples, we account for the contribution of a feature either to the similarity but also to dissimilarity. We remark, that such a novel technique enables the calculation of feature importance, where in our case each feature corresponds to some gene-associated information, such as gene expression or genetic mutations (see Table 3), specifically for each of the 24 cancer types described in Table 2. This represents a significant advancement over the system proposed in [17], where feature importance was assessed across the entire dataset rather than within specific classes.

Results

We start with a description of the data used for experimentation, including the preprocessing steps employed to prepare the data for analysis. Following, we provide details about the the experiments carried out to assess the effectiveness of the proposed SNN (S). Finally, we describe the insights and the cancer-related knowledge gained by explaining the trained (S) model behavior using the SHAP values-based technique proposed described the *Methods* section.

Data acquisition and preprocessing

The *TCGA* program has extensively characterized over 20,000 primary cancer and matched normal samples across 33 different cancer types. For each analyzed patient (*sample*), a comprehensive set of *genetic*, *omics*, and *clinical* data is available in a separate files. In order to build the dataset used for our experiments, after downloading the whole dataset, we only retained samples for which information about *genomic mutations* and *gene expression* profiles were available. We recall that, with the term *genomic mutations* we generically refer to: *(i) Single Nucleotide Polymorphism* (SNP), *insertions* (INS), and *deletions* (DEL) occurring on the genes of a given subject; for each patient, this information was summarized by counting the frequency of each type of mutation occurring in each gene; *(ii) Copy Number Aberrations* (CNAs) values, which were normalized such that each gene could have a copy number value of 0, 0.25, 0.5, 0.75 or 1. We then selected only those subjects diagnosed with tumor types that had a sufficient number of samples to effectively leverage the available data and the capabilities of the developed SNN. Various minimum thresholds were tested, but best results were achieved by setting the minimum sample size per cancer type to 9. Following the application of these filters, our dataset included 8,788 patients across 24 distinct cancer types, with data available for 23,665 genes per patient (see Table 2 for a detailed overview of the 24 selected cancer types and the corresponding number of patients for each).

Table 2. Selected Cancer Types and Number of Patient Samples

ID	Cancer Type	# Patient samples
1	*Adrenocortical Carcinoma*	76
2	*Bladder Cancer*	401
3	*Breast Cancer*	1045
4	*Cervical Cancer*	289
5	*Colorectal*	571
6	*Endometrial Cancer*	563
7	*Esophagogastric Cancer*	589
8	*Glioma*	607
9	*Head and Neck Cancer*	495
10	*Hepatobiliary Cancer*	392
11	*Leukemia*	132
12	*Mesothelioma*	78
13	*Miscellaneous Neuroepithelial Tumor*	22
14	*Non-Hodgkin Lymphoma*	48
15	*Non-Small Cell Lung Cancer*	977
16	*Ocular Melanoma*	53
17	*Ovarian Epithelial Tumor*	200
18	*Pancreatic Cancer*	159
19	*Pheochromocytoma*	102
20	*Prostate Cancer*	469
21	*Renal Cell Carcinoma*	838
22	*Soft Tissue Sarcoma*	234
23	*Thymic Tumor*	106
24	*Thyroid Cancer*	342

This dataset presents a classic example of the "curse of dimensionality" in bioinformatics, characterized by high-dimensional data with a relatively low number of samples. To address this, we reduced the number of genes considered by filtering based on gene expression levels, following a similar approach proposed in [17]. First, we applied the *min-max normalization* to normalize the gene expression across all the genes in our dataset. Following, for each gene, we calculated the standard deviation of its expression across all cancer types. Subsequently, we retained only those genes with a standard deviation above a specified threshold. As we can see in Figure 2, by setting the standard deviation to 0.0005 we can consider approximately 90% of the genes, and so we selected such a value as reference threshold. This filtering resulted in a subsets of 1,349 *selected genes*. Following, we considered two different datasets for our experiments: *(i)* one considering uniquely the gene expression of each one of the 1,349 genes (1,349 features), and *(ii)* the other considering also the frequency of each type of mutation occurring in each of the 1,349 genes, for a total of 5,719 features obtained as follows: we considered the 1,349 features of the gene expression, and for each of such value (one gene) we also consider 4 additional information regarding INS, DEL SNP, and CNAs, for a total of $1,349 \times 5 = 6,745$; to reduce the sparsity of the dataset, we decided to remove all columns related to genomic mutations that

had a value of 0 across all samples, resulting in the elimination of 1,026 columns; so, the final number of features considered is $6,745 - 1,026 = 5,719$. Table 3 summarizes the number of the features considered for each sample into the two datasets produced.

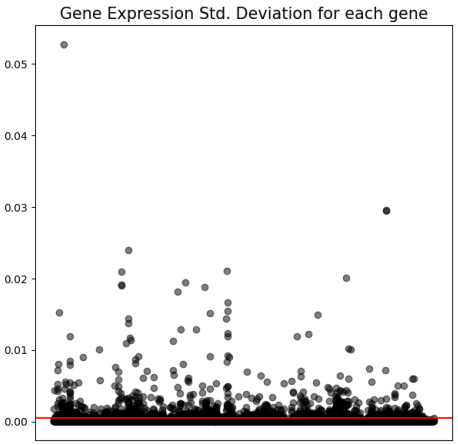


Figure 2. For each gene, we calculated the standard deviation of its expression across all cancer types. Following, we selected only those genes with a standard deviation higher than 0.0005. The chosen threshold is displayed as a red line in the plot.

Table 3. Summary statistics for the two datasets, *Only Gene Expression* (GE) and *Gene Expression + Mutations* (GEM), produced for the experiments, each one with a total number of 8,788 samples (patients).

Dataset	# Samples	# Genes	# Features
Only Gene Expression (GE)	8,788	1,349	1,349
Gene Expression + Mutations (GEM)	8,788	1,349	5,719

Given a specific dataset (GE or GEM summarized in Table 3), and given a specific patient sample p in such a dataset, we will indicate with $fv(p)$ its vector of features, or *feature vector* (vector of size 1,349 for the dataset GE and of size 5,719 for the dataset GEM).

Experiments

We recall that, as explained above, in this work, we extend *CancerSiamese* [17]. However, we will demonstrate that, to address the research questions outlined in the *Methods* section, the structure of *CancerSiamese* is insufficient and does not fully exploit the capabilities of SNNs to integrate both gene expression and genetic mutations. Moreover, *CancerSiamese* entirely overlooks the potential to combine expression data with mutational profiles to identify a set of type-agnostic marker genes, whose combined information can define the similarity or dissimilarity between samples from the same or different TMEs of cancer types. To overcome these limitations, we have designed our experiments to use *CancerSiamese* as a baseline for comparison across the datasets described in Table 3.

Specifically, the experiments have been organized as follows:

- E1. *CancerSiamese* C on GE dataset: as a baseline, the SNN architecture of *CancerSiamese*, denoted with C, was trained on the *Only Gene Expression* (GE) dataset, consisting of gene expression information, to identify 24 types of cancer detailed in Table 2;
- E2. *CancerSiamese* C on GEM dataset: the SNN model C was also trained on the *Gene Expression + Mutations* (GEM) dataset, consisting of both gene expression and genetic mutations information data, to identify 24 types of cancer detailed in Table 2;
- E3. Proposed SNN S on GEM dataset: the proposed SNN model architecture described in Figure 1, and denoted with S, was trained on the *Gene Expression + Mutations* (GEM) dataset, consisting of both gene expression and genetic mutations information data, to identify 24 types of cancer detailed in Table 2;

For each experiment, both the SNNs S and C were trained on one of the training dataset described in Table 3 (refer to the *Methods* section for details) with *Keras DL* platform with the *Tensorflow* backend. Specifically, for the experiments E1 and E2, both S and C were trained on the GE dataset, while the experiments E3, both S and C were trained on the GEM dataset. As explained in *Methods* section, for each experiment, network transfer learning was conducted from the training of both S and C models, where the initial weights of the 1D-CNN feature extractors were set as those in the 1D-CNN for classification of cancer types pretrained on the same training set. The weights for the rest of the layers (i.e. FC and sigmoid) were initialized by Xavier Initialization as also proposed by [17]. Each SNN network was optimized with a binary cross-entropy loss and trained with 20,000 training iterations, where each iteration includes a batch of 128 pairs with an equal number of matched and mismatched pairs, all chosen randomly from the corresponding training dataset. The network parameters were optimized by Adam optimizer and all of the hyperparameters were tuned manually and summarized in Appendix. For each experiment, the trained networks were tested for different N-way predictions, but best results have been obtained with $N = 24$. For an N-way prediction, each SSN compared a query sample with a support set of N samples, each from a different cancer type. The type of the query sample was predicted as the type of the paired support sample if the pair received the highest probability by the SNN out of the N pairs. The prediction was counted as correct if the predicted type was the same as the true type of the query sample. For every N-way prediction, we tested each SNN on 20,000 randomly selected query samples and the corresponding support set from the test dataset and each support set contained N randomly selected samples, each from a different cancer type but one of them coming from the same cancer type as the query sample. The *TCGA* datasets used to construct the primary dataset in this study can be accessed via the *cBioPortal for Cancer Genomics* platform[2]. The datasets produced and used for the experiments[3], and the developed source code are available online[4].

As for the experiment E1, when we use the baseline SNN C on the GE dataset, we obtained a higher accuracy (91%) than the highest one obtained by the authors in the original paper [17] (89.67%) for the same task on the same type of data. Such discrepancy can be explained by having slightly different data. However, we highlight that our data comprised a higher number of cancer types (24 against 19), and we evaluated the accuracy achieved by C using a 24-way (against the 10-way used by [17]. As for the experiment E2, when we use the baseline SNN C on the GEM dataset, that is, on a more complex input obtained by adding genetic mutations to the gene expression, the architecture of C is shown to be not powerful enough to handle both gene expression and gene mutations simultaneously, achieving accuracy 22.4%. This was expected, since we added a a high number of features by considering the mutations, while leaving the number of samples unchanged. In order to overcome this limitations, we ran the experiment again using the deeper architecture of S described in Figure 1. With this setting, and using a 24-way to evaluate the model's performance, we were able to achieve an accuracy higher than C equal to 85.3%, including also the mutational information. A summary of the results obtained through the different experiments can be seen in Table 4.

2 https://www.cbioportal.org/
3 https://drive.google.com/drive/folders/1_mIUXzWdfXwy4XapOQedXiA6G9aEutZQ?usp=drive_link
4 https://github.com/FLaTNNBio/SNN_for_Enhanced_Cancer_Type_Detection

Table 4. Results achieved by the three different experiments settings

Model	Dataset	# Cancer types (classes)	# Subjects (rows)	# features	Accuracy
C *baseline* architecture	GE	24	8,788	1,349	91.1%
	GEM	24	8,788	5,719	22.4%
Proposed SNN S	GEM	24	8,788	5,719	85.3%

Cancer Insights from SHAP-Based Explainability

As discussed above, we conducted our experiments using the mutational data of each gene, which, while TMB is traditionally defined as the number of somatic mutations per megabase of genomic sequence [57], aligns more closely with the interpretation in [58], where TMB is treated as the total number of mutations per gene, similar to our approach. After training the proposed S on these features (experiment E3), we applied the novel SHAP-based technique described in the *Methods* section to assign an *importance* score to each feature according to the cancer type. Our objective was to identify the features that were most discriminative for each cancer type, i.e., the features that enabled the model to distinguish a specific cancer type from others. Since SHAP values are additive, we aggregated the feature importances by gene, summing the contributions from its *expression*, *SNPs*, *INS*, *DEL*, and *CNAs*. Then, we ranked the genes based on their overall importance. For instance, Figure 3 shows the contribution of each aggregated feature for the top 20 most important genes in *Pancreatic Cancer*. A more detailed analysis can be found at[5], where we report the importance of the top 50 genes for each of the 24 cancer types, highlighting the contribution of each one of the aggregated features.

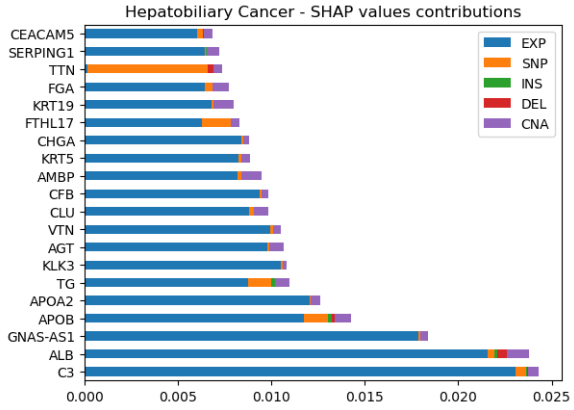


Figure 3. Top 20 most important genes in *Pancreatic Cancer*

After identifying the top 100 genes for each cancer type, we analyzed how many times each one of these genes is present in the top 100 genes of "how many" cancer types, i.e., we checked how many of these 100 genes were in the top 100 for one cancer type, then 2, then 3, and so on, up to 24, which is the maximum number of cancer types analyzed. As we can see in Figure 4, on a total amount of 404 distinct genes, 168 appear in a single cancer type, while 36 genes are consistently marked as important across all the 24 cancer types. We will refer to this set of genes as *shared genes*. Then, we carried out three different analysis: *(i)* enrichment analysis of the 48 genes appearing at least in 20 cancer types (see Figure 4), *(ii)* analysis of the *shared genes*, i.e., those 36 genes which are important for all 24 cancer types, and *(iii)* analysis of the *specific genes*, i.e., those genes which were considered important only for a subset of the 24 cancer types or even just only one cancer type.

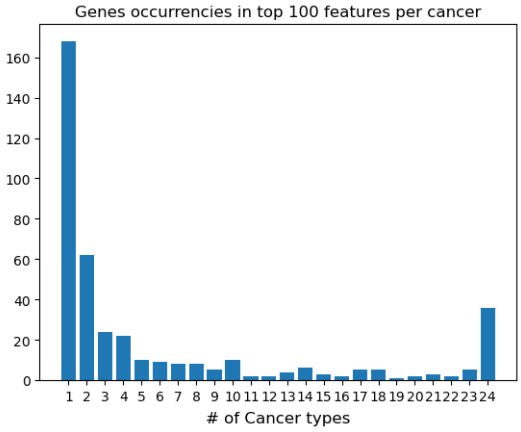


Figure 4. Presence statistics of 100 top genes in "how many" cancer types

Enrichment analysis. To enhance the significance of our enrichment analysis and to capture genes that may have been excluded due to fluctuations in SHAP value calculations, we opted to lower the cutoff threshold to at least 20 cancer types out of 24. This adjustment allowed us to expand the subset of genes, considered in the analysis, to 48 genes. On such genes, we performed an enrichment analysis using *The Database for Annotation, Visualization and Integrated Discovery* (DAVID), which is an integrated biological knowledge-base and analytic tools aimed at extracting biological meaning from large gene/protein lists [59]. We found that these genes are enriched for components related to the *TME*, such as proteins and elements found in extracellular matrix (ECM). This is consistent with what reported by [17]. As discussed in *Introduction* section, it is known in literature that different cancer types are characterized by different TMEs, which can also determine the vulnerability of the cancer to different therapies [60]. TMEs are defined as normal tissues surrounding the tumor which are shaped by it, and which can determine several things, such as its spreading. This means that different TMEs are usually made by the same types of components (such as blood vessels, extra cellular matrix components) but performing different functions among the cancers. Details about this analysis can be found in Table 5. Additional materials produced during our analysis can be found online[6].

Shared genes analysis. Since each tumor possesses a TME composed of similar components, such as healthy tissues that are "co-opted" by the tumor, yet the functional behavior of the TME, including its immune responsiveness and biological roles, varies across cancer types, it follows that there may be a set of TME-related genes considered important for all cancers. Given this, it is reasonable to assume that the proposed SNN, by focusing on these genes, can learn to distinguish between TMEs, allowing it to differentiate various cancer types effectively. This can be see in detail in Figure 10, where we can observe that the same genes, although being among the most important genes for all the different cancer types, can assume highly diverse importance values. As we can see, the importance of these genes is mainly given by their *expression*. This is not surprising, since gene expression was the primary criteria used to select the subset of genes to be analyzed. However, as we can see in Figure 11, the distribution of importances of the TMB among *shared genes* varies among the different cancer types, contributing to the overall result. This leads us to believe that that different mutational patterns may contribute to the TMEs varieties in functions.

Several *shared genes* identified as important from the SHAP values based on our trained model, nicely relate to several known re-

5 https://drive.google.com/drive/folders/10dZBJvN4WKJ4tul61-Au_giUNssfHwEB?usp=drive_link

6 https://drive.google.com/file/d/1D8sOmleGA-XEQiC8uhg7yMnmyF4cTQmh/view?usp=drive_link

Table 5. Annotation Clusters with Enrichment Scores and Corresponding Statistics

Annotation Cluster	Enrichment Score	Annotations	Count	P-value	Benjamini
Cluster 1	7.06	*extracellular space*	26	9.9E-15	7.7E-13
		extracellular region	22	2.1E-10	1.1E-8
		Secreted	22	1.3E-9	3.2E-8
		Signal	24	7.3E-6	8.8E-5
		Glycoprotein	26	1.0E-4	1.3E-3
		Disulfide bond	23	1.3E-4	1.3E-3
		N-linked (GlcNAc...) asparagine	21	1.5E-4	6.2E-2
Cluster 2	3.46	*collagen-containing extracellular matrix*	9	1.9E-6	5.9E-5
		Extracellular matrix	7	3.8E-5	4.8E-4
		extracellular matrix structural constituent	5	1.3E-4	5.9E-3
		extracellular matrix	5	1.6E-3	2.3E-2
		extracellular matrix organization	4	6.7E-3	5.0E-1
		basement membrane	3	1.8E-2	1.6E-1
Cluster 3	2.64	*Cytoskeleton in muscle cells*	8	2.5E-6	2.3E-4
		protease binding	6	6.3E-6	1.1E-3
		collagen binding	5	1.7E-5	1.5E-3
		ECM-receptor interaction	5	1.0E-4	4.7E-3
		endoplasmic reticulum lumen	6	4.8E-4	1.0E-2
		Human papillomavirus infection	5	1.3E-2	2.9E-1
		Cell attachment site	3	1.4E-2	5.6E-1
Cluster 4	2.34	*proteinaceous extracellular matrix*	7	2.3E-5	2.1E-3
		extracellular matrix	7	3.8E-5	2.2E-3
		glycoprotein	7	7.4E-5	1.8E-2
Cluster 5	2.16	*heparin-binding*	6	6.8E-5	2.7E-3
		ECM-receptor interaction	6	1.1E-4	3.2E-3
		Cell adhesion molecules	6	3.0E-4	7.9E-3
Cluster 6	1.98	*Proteoglycans in cancer*	5	1.9E-2	2.9E-1
		PI3K-Akt signaling pathway	5	1.8E-2	2.9E-1
Cluster 7	1.82	*Disulfide bond*	3	5.2E-2	3.8E-1
		Focal adhesion	3	5.0E-2	4.5E-1
Cluster 8	1.66	*Cell adhesion*	3	1.0E-1	6.3E-1
		Glycoprotein	3	1.3E-1	7.4E-1
Cluster 9	1.50	*basement membrane*	2	1.2E-1	8.2E-1
Cluster 10	1.36	*Motif: Cell attachment* site	2	1.4E-1	1.0E0
Cluster 11	1.20	*Secreted*	1	2.3E-1	1.0E0
		Signal	1	2.3E-1	1.0E0

sults in literature. This is an additional capability we have compared to *CancerSiamese*. Indeed, while *CancerSiamese* highlighted some genes that could potentially serve as general biomarkers (meaning they are important for the network overall), we can identify genes that are more or less significant for specific cancer types, such as those listed below. For example, we can see that the *KLKL3* seems to be extremely important when it comes to the *Prostate Cancer* (see Figure 10). This looks coherent with the literature, which reports that such a gene is in fact a known prostate cancer biomarker [61, 62]. A similar event can be observed for the *Thyroid Cancer*, with the *TG* being the most important gene. Also in this case, such gene is a well established thyroid cancer biomarker [63]. Similarly, the *KRT5* has been reported to be highly expressed in the *Head* and *Neck Cancer* [64], and *CEACAM5* has been identified as a biomarker for *Colorectal Cancer* [65], among others. For example, genes from the *Collagen VI family* (*COL6A*) are a major member of extracellular matrix protein, which is involved in tumor genesis and tumor progression, and they are differently expressed across different cancers [66]. In figure 11 we can observe that the presence of CNAs on the *COL6A4P1* gene seems particular important in the context of the *Bladder Cancer*, and it would be worthy of further investigations. Such observations seems in agreement with the recent finding that *Bladder Cancer* is, in general, affected by a high number of CNAs [67]. The broad impact of CNAs on multiple genes in bladder cancer is evident in our results, as can be seen in, for example, in Figure 5. Another noticeable example is the high *GNAS-AS1* importance in the *Adenocortical Cancer*. Such gene was reported to be up-regulated compared to a healthy control group [68].

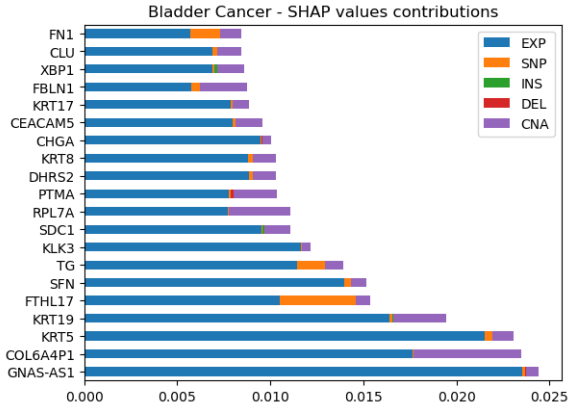


Figure 5. SHAP values for the 20 most important genes in Bladder Cancer.

To assess the role of *shared genes* in a cohesive manner, we analyzed them function through the *STRING* platform [69], which is a database of known and predicted protein-protein interactions useful for Protein-Protein Interaction Networks Functional Enrichment Analysis. As a result, 18 out of the 36 shared genes were clustered together and classified as *tumor markers*. Figure 6 shows the connections among the members of the clusters, where a higher number of connections suggests a higher reliability of the data, since it corresponds to a higher number of findings in literature.

Overall, the literature supporting our findings suggests that *shared genes* considered important for which there are not known

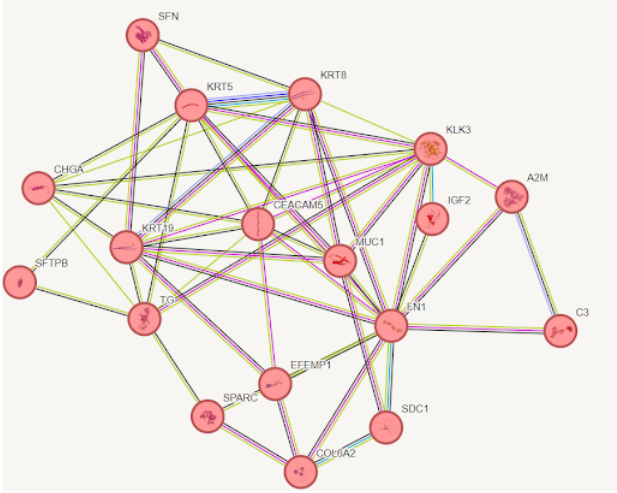


Figure 6. Cluster detail of 18 genes reported as tumor biomarkers by the *STRING*.

biological results may be valuable of deeper analysis. Therefore, beyond being a valuable diagnostic tool, our method could be a potential tool to select interesting targets of study in cancer.

Specific genes analysis. Among the genes considered important only in some (or even just one) cancer types (*specific genes*), several genes were chosen mainly due their mutational patterns. As explained above, all the plots showing the most 50 important genes for each cancer type studied can be found online. It is known that cancer development tends to generate a high number of mutations which do not actually give any evolutionary advantage [60]. This makes it difficult to identify some *signature mutations*, i.e., mutations which are specific to a cancer type and that may have an actual functional role in its development. However, we were able to identify some important genes chosen for their mutational patterns which are supported in literature. This suggests that our method may be a valuable tool for directing the attention of biologist on a promising subset of genes when looking for untactful mutations.

In Figure 7, we can see that, as for the *Leukemia*, gene *NPM1* importance was mainly determined by the frequency of the *insertions* mutation. We remark that, it is know in literature that insertions in the *NPM1* gene are present in the 30-50% of patients affected by acute *Myeloid leukemia* [70]. When we looked into our data, we found out that 41 patients out of 132 affected by *Leukemia* patients presented one insertion on the *NPM1*, while no other patient affected by another cancer type had any insertion on this gene.

Similarly, in the *Endometrial* cancer (*EC*), we can observe that the *DST* gene was chosen mainly due to the impact of its SNPs mutation frequency (Figure 8). [71] carried out a preliminary study on a cohort of 23 patients affected by *uterine sarcomas*, *carcinosarcomas*, *endometrial stromal sarcoma* (*ESS*), *adenosarcoma*, and *leiomyoma*. One of their goal was to identify some *signature genes*. Among other results, they classified the SNP mutation (16429C>T) on the *DST* gene as a *signature gene* for the *ESS* cancer (i.e. our *Endometrial cancer*). However, in their cohort only 7 patients were affected by *ESS*, and only 2 of them shown the aforementioned mutation. Our model seems to confirm the importance of SNPs on the *DST* gene for the *ESS* cancer type, even using a much bigger cohort, since our dataset included 563 *ESS* patients. Overall, 10 among the top 30 predicted genes with respect to the endometrial cancer find confirmation in literature [72, 73, 74, 75, 76, 77, 78, 79, 80, 81]. Specifically, *TUBB4B* gene is a known prognostic marker in endometrial cancers [72], and *SERPINC I* is reported as an important blood-based protein biomarker candidates for *EC* detection [73]. Similarly, *RPL7A* expression was associated to the Uterine Corpus Endometrial Carcinoma [74]. Genes such *SPOCK2*, *XBP1*, *SPARC* and *SDC1* are connected to *EC* growth, progression and migration [75, 76, 77, 78]. Findings about *MALAT1* suggested that the *rs664589 C>G* mutation significantly increased the risk of endometrial cancer [79]. Interestingly, mutations are responsible for a considerable portion of the overall importance assigned to the *MALAT1* gene by our model for the *EC*, although multiple type of mutations are involved, as can be seen in Figure 8. A final valuable finding is that *CTNNB1* appears to be a frequently mutated gene in, with its mutations leading to alterations in the Wnt/β-catenin signaling pathway, which is involved in the carcinogenesis and progression [82]. Such gene is among the top 100 most important genes for *EC* according to our model and, remarkably, most of its importance is determined by its mutations, coherently with the aforementioned literature. This can be observed in the additional material available online [7].

Another interesting case is the *Non-Small Cell Lung* cancer (*NSCLC*) one. As can be seen in Figure 9, several genes were marked as *important* due to the impact of their mutational patterns. In particular, the *TFRC*, *CTNNB1*, *TTN*, *SDC2*, *PLEC*, *AP2M1*, *COL4A1* and *AEBP1*. Some of these genes have some confirmation in literature. For example, it is reported that oncogenic mutations have been found on *CTNNB1* in liver cancer, uterine cancers, and in a small subset of NSCLCs, although but clinical significance of these CTNNB1 gene mutations in *NSCLCs* and *Lung Adenocarcinomas* (*LUADs*) is still unclear [83]. Similarly, it is reported that *TTN* is a common mutated gene in diverse types of tumor including *LUAD* and *lung squamous cell carcinoma* [84]. In [85], the authors suggest that *SDC2* likely plays an important role in the invasive properties of *LUAD* cells. These findings may suggest that genes marked as important by the model may be worthy of further investigations.

SNN evaluation for *Healthy* vs. *Cancerous* detection

As final experiment, we evaluated the capability of the proposed SNN S, trained on the GEM dataset, to distinguish between *healthy* and *cancerous* samples. This analysis aimed to investigate whether S could be applied as a reliable "binary" classifier to ascertain a patient's health status. To this end, we constructed a dataset of 305 healthy individuals using data from the *TCGA* project, accessed through the *cBioPortal for Cancer Genomics* platform, ensuring that each individual was characterized by the same 5,719 features, i.e., *Gene Expressions + Mutations* (see GEM dataset in Table 3). This enabled direct application of the trained S without modification.

To formalize the similarity criterion for assessing whether a healthy patient p_i belongs to a given cancer class C_l, we proceed as follows. Let mean_l be the mean similarity score for C_l, previously calculated using S on the pairs of samples from this class. We denote by $\{r_{l,1}, \ldots, r_{l,k}\}$ a set of k randomly selected *representative* samples from class C_l. The similarity score between p_i and $r_{l,j}$ with $j = 1, \ldots, k$, is calculated using the trained SNN S as follows:

$$d(p_i, r_{l,j}) = \texttt{L2}\left(\texttt{S_q}(fv(p_i)), \texttt{S_s}(fv(r_{l,j}))\right)$$

where L2 denotes the Euclidean distance between the embeddings generated by the query subnetwork S_q and the support subnetwork S_s (see *The proposed SNN* section). Let $S_i(C_l)$ be the set of representatives samples within $\{r_{l,1}, \ldots, r_{l,k}\}$ "similar" to p_i:

$$S_i(C_l) = \{r_j \in C_l \mid \text{mean}_l - \epsilon \leq d(p_i, r_j) \leq \text{mean}_l + \epsilon\}$$

where the parameter ϵ defines the *similarity tolerance margin*, acting as a control for the interval $[\text{mean}_l - \epsilon, \text{mean}_l + \epsilon]$. It specifies the acceptable range of deviation from mean_l. A smaller ϵ implies a stricter similarity criterion, increasing specificity, while a larger ϵ broadens the acceptance range, allowing for greater variability in

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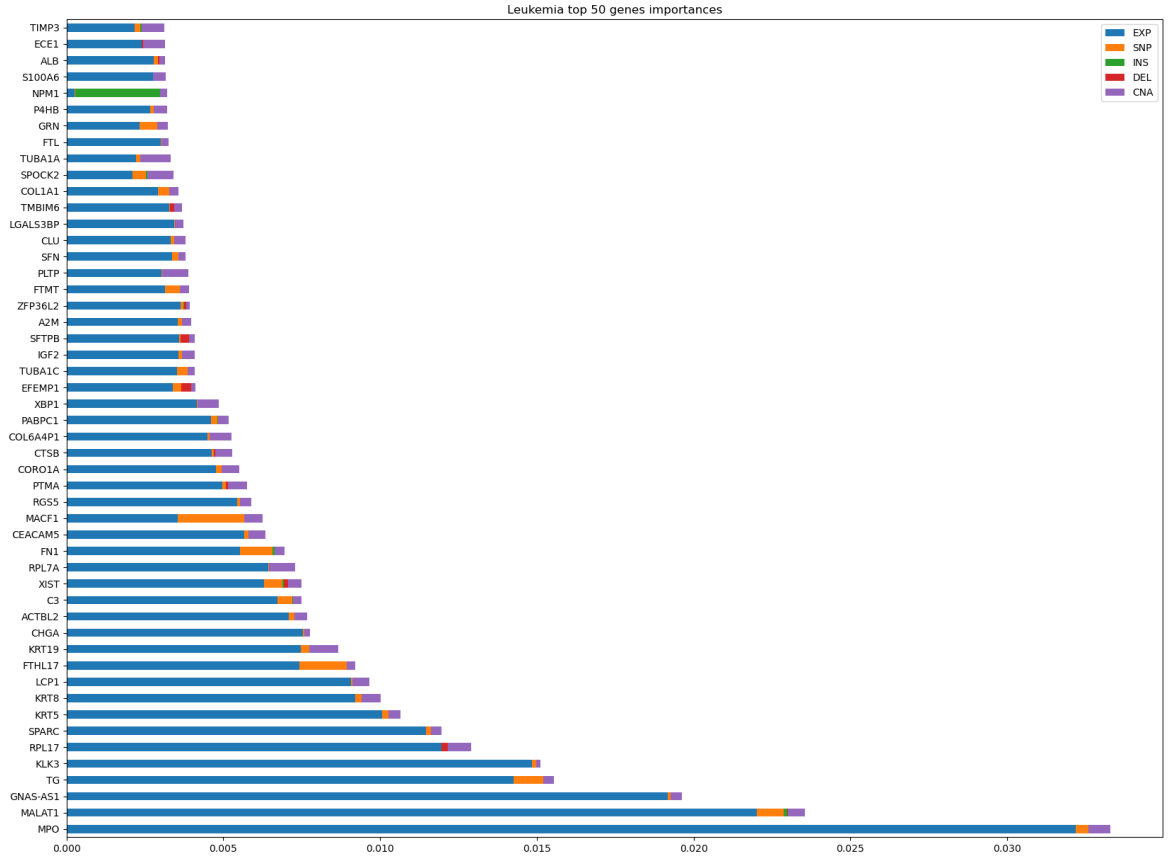


Figure 7. Shap values for the 50 most important genes for Leukemia.

the similarity without compromising classification robustness.

The patient p_i is classified as likely belonging to C_l if the *majority* of the selected representatives satisfy the similarity condition, that is, $|S_i(C_l)| \geq \frac{k}{2}$. On the other hand, if the number of representatives satisfying the similarity condition is less than half, i.e., $|S_i(C_l)| < \frac{k}{2}$, then the patient p_i is classified as not belonging to C_l, implying that p_i is considered *healthy* with respect to C_l. This formulation provides a robust method for assessing the health status of p_i by comparing it against multiple representatives of each cancer class, leveraging the learned similarity function of S for a reliable classification decision.

Through extensive experimentation with varying values of k and ϵ, we determined that setting $k = 3$ and $\epsilon = 0.03$ yielded best results. Under these parameters, the system accurately identified 93% of the healthy patients as non-cancerous. To further validate the robustness of this approach, we conducted a similar experiment on 305 randomly selected cancerous patients from the GEM dataset. Using the same classification criterion, S accurately identified 99.3% of the cancerous patients as belonging to a cancer class, thus affirming its effectiveness in recognizing malignancy.

These results underscore the S's stability and robustness, demonstrating its ability to perform binary classification, *healthy* vs. *cancerous*, without the need for retraining, and on healthy patients never seen during the training. The use of multiple class representatives k enhances the model's reliability, as it ensures that classification decisions are based on a broader sampling from each cancer class. This approach has clear implications for practical applications, where testing across multiple representatives strengthens the model's response consistency and reduces the likelihood of misclassification, making S a viable tool for clinical diagnostics.

Conclusions and Discussions

This study presents a novel approach to cancer classification using *Siamese Neural Networks* (SNNs), integrating gene expression data with genetic mutations, across multiple cancer types. The method aims to tackle two key challenges in cancer genomics: the inherent heterogeneity of tumors and the limitations of using "single-omic" data (such as gene expression or mutation data alone) for classification tasks. Tumors from the same tissue type can exhibit vastly different genetic profiles due to mutations, alterations in gene expression, and varying interactions with the *tumor microenvironment* (TME). This complexity makes it difficult for traditional models that rely on either mutations or gene expression alone to fully capture the tumor's characteristics. We address this by integrating gene expression profiles and genetic mutation data, thereby capturing complementary aspects of tumor biology that are missed when these data are analyzed in isolation. By integrating both gene expression or mutation data, our model can capture the underlying biological differences that drive the diversity within cancer subtypes, resulting in a more nuanced and accurate classification.

Explainable insights. The SHAP-based explainability analysis provided crucial biological insights, particularly highlighting the ability of our model to identify genes that distinguish various cancer types. One key finding was the identification of 36 shared genes consistently important across 24 cancer types. Many of these genes are linked to the *tumor microenvironment* (*TME*), which plays a critical role in cancer progression and metastasis. For example, the *COL6A4P1* gene emerged as particularly important in *Bladder Cancer*, where *CNAs* play a significant role in tumor evolution. This finding aligns with recent research that emphasizes the prevalence of *CNAs* in bladder cancer, further suggesting their potential as biomarkers for early detection or therapeutic targeting. Our analysis also

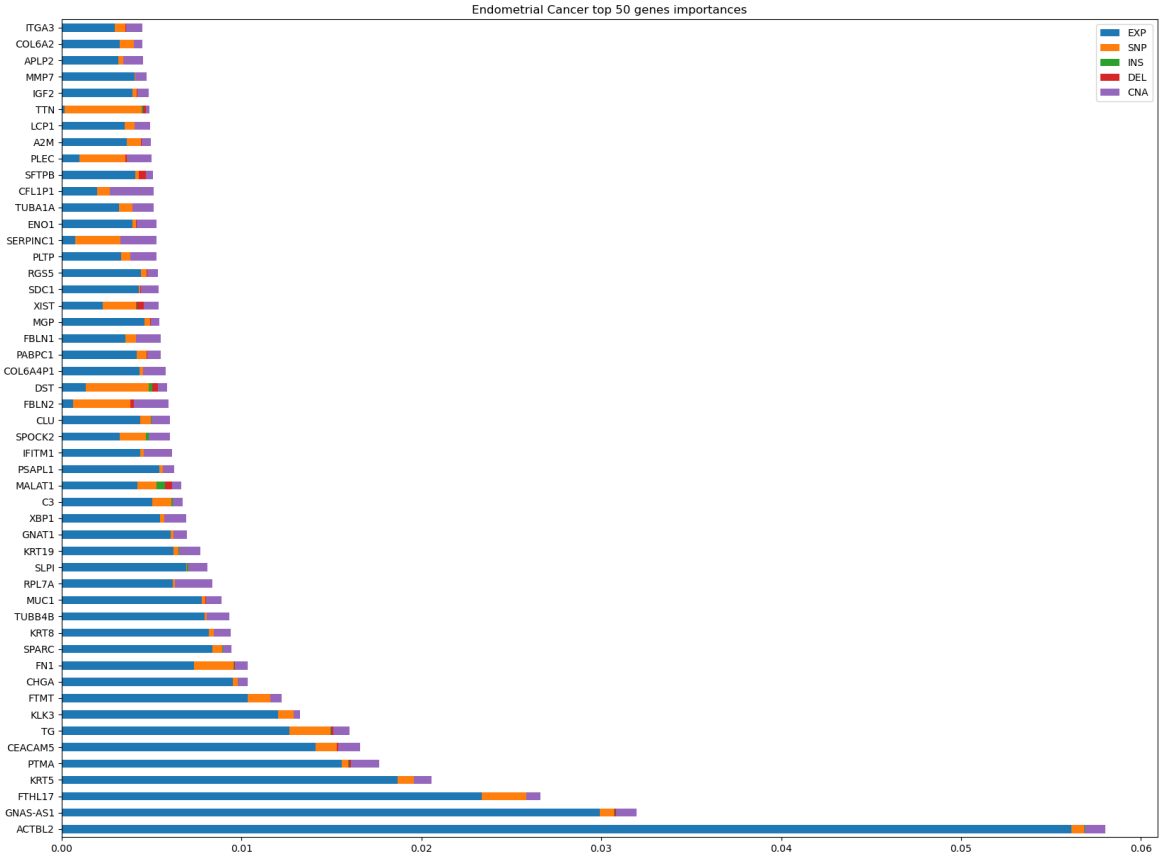


Figure 8. Shap values for the 50 most important genes for Endometrial Cancer.

highlighted individual *shared genes* that are already recognized as cancer biomarkers. For instance, *KLKL3* was identified as a top-ranking gene in *Prostate Cancer*, consistent with its established role as a prostate-specific antigen used in clinical diagnostics. Similarly, *TG* (*Thyroglobulin*) was found to be a key gene in *Thyroid Cancer*, aligning with its clinical use in monitoring post-treatment recurrence. Another gene, *KRT5*, emerged as significant in *Head and Neck Cancer*, especially in squamous cell carcinomas, while *CEACAM5* was flagged for its role in *Colorectal Cancer*, where it is widely used as a marker for monitoring disease progression and metastasis. Beyond shared genes, the model also identified specific genes that play crucial roles in individual cancers. For example, in *Acute Myeloid Leukemia* (*AML*), the *NPM1* gene was significant due to the presence of insertion mutations. Our data showed that 41 out of 132 *AML* patients had these insertions, highlighting *NPM1* as a key driver of *AML* pathogenesis and showcases the model's capacity to pinpoint critical mutations that are essential in specific cancers. Similarly, in *Endometrial Cancer* (*EC*), the *DST* gene was identified as important due to its *SNP* mutations. Literature has suggested that *DST* mutations are associated with specific subtypes of endometrial cancer, particularly *endometrial stromal sarcomas* (*ESS*). Our findings, based on a large cohort of 563 *ESS* patients, further validate *DST* as a potential biomarker for this cancer subtype, highlighting its clinical relevance. In *Non-Small Cell Lung Cancer* (*NSCLC*), several genes were flagged for their mutational patterns, notably *CTNNB1* and *TTN*. *CTNNB1*, often implicated in other cancers like liver and uterine cancers, was found to have a significant role in *Lung Adenocarcinoma* (*LUAD*). While its function in *NSCLC* is still under investigation, this finding opens new avenues for research into *CTNNB1*'s role in lung cancer biology. The SHAP-based analysis showed that while gene expression was a key factor, mutational patterns such as SNPs, insertions, and deletions also played a significant role in distinguishing cancer types, highlighting the value of multi-omic methods in uncovering genetic mechanisms, new biomarkers and therapeutic targets in cancer research.

Strengths, limitations, and future directions. In addition to integrating gene expression and mutation data for improved classification and a more comprehensive view of tumor biology, one of the key strengths of using SNNs is their ability to effectively learn from limited and imbalanced datasets, which is a common challenge in cancer genomics. SNNs leverage pairwise comparison learning, making them particularly suited for scenarios where high-quality labeled data is scarce. This method allows the model to learn discriminative features from fewer examples by focusing on similarity metrics between pairs, thus reducing the dependency on large, labeled datasets typically required by traditional machine learning models. In the context of cancer genomics, this is particularly relevant, as acquiring extensive labeled data, especially for rare cancer types or subtypes, can be difficult. Furthermore, SNNs mitigate the risk of overfitting, as the model is trained to differentiate between subtle biological differences, such as those in gene expression and mutation patterns across tumor types, rather than memorizing individual samples. This ability to generalize well from limited data is crucial for building robust models in precision oncology.

However, this study is not without limitations. First, by choosing the variance in the gene expression as criteria for filtering, we potentially filtered out genes which may carry important mutations while not being differentially expressed. This suggests that investigating other filtering strategies could be a valuable future development. Secondly, while the results are promising, the model was developed and tested on high-quality tumor tissue samples. In real-world clinical settings, such samples may not always be available. Moving forward, applying the model to more accessible data types, such as those obtained from liquid biopsies (e.g., circulating tumor DNA), would be an important step. However, liquid biopsy

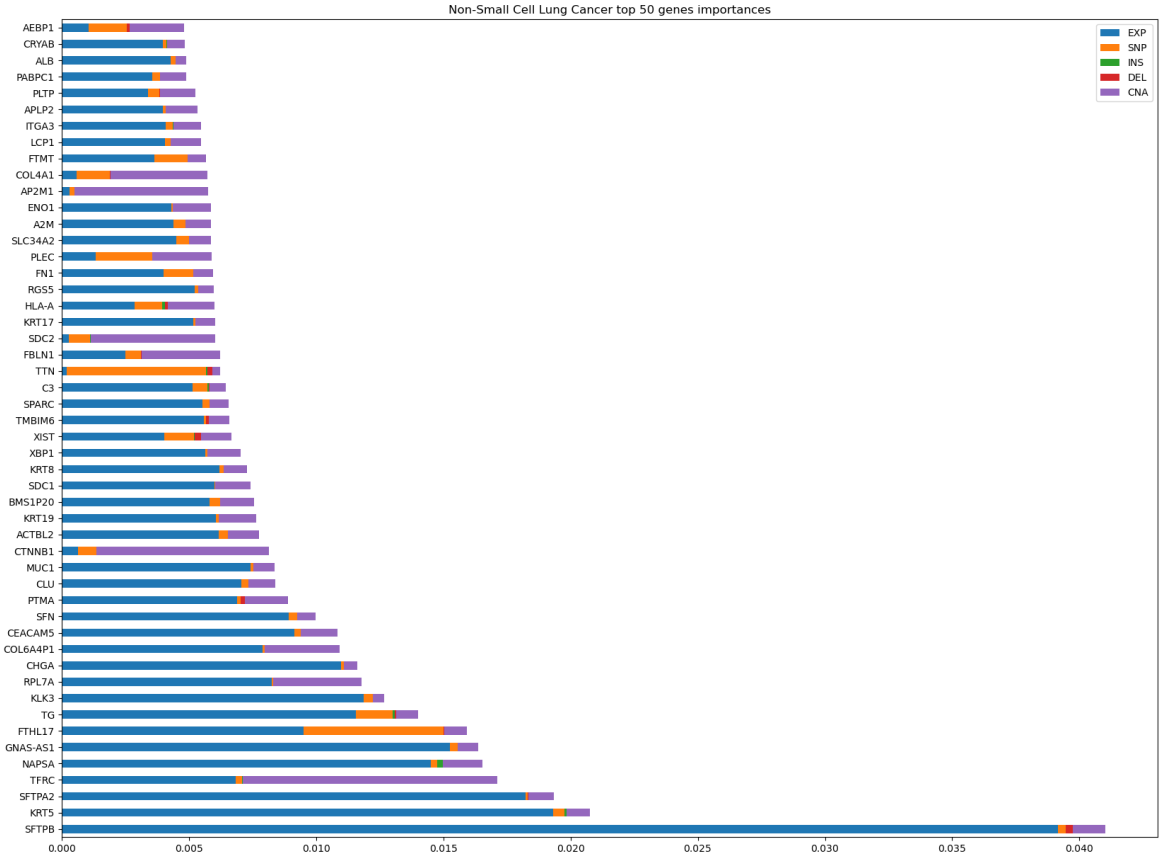


Figure 9. Shap values for the 50 most important genes for Non-Small Cell Lung Cancer.

data is often noisier and presents challenges such as lower resolution and the dilution of tumor-derived DNA in the bloodstream, which could impact model performance. Additionally, the model currently relies solely on genomic and transcriptomic data, but future iterations could incorporate proteomic and metabolomic data to capture even more aspects of tumor biology.

Future work should focus on adapting the model for use with more clinically feasible data sources, such as liquid biopsies. This would make the method more applicable in routine clinical diagnostics, enabling non-invasive, real-time monitoring of cancer progression. Moreover, expanding the model to integrate other layers of biological data (such as proteomics or metabolomics) could further enhance its ability to capture the full complexity of cancer biology. Lastly, applying the model to early detection and prediction of precancerous conditions could open up new avenues for improving cancer survival rates through earlier intervention.

Availability of source code and requirements

Lists the following:

- Project name: *SNN_for_Enhanced_Cancer_Type_Detection*
- Project home page: e.g. `https://github.com/FLaTNNBio/SNN_for_Enhanced_Cancer_Type_Detection`
- Operating system(s): *Platform independent*
- License: *MIT license*.

Data availability

The *TCGA* datasets used to construct the primary dataset in this study can be accessed via the *cBioPortal for Cancer Genomics platform* at `https://www.cbioportal.org/`.

The datasets produced and used for the experiments can be downloaded from `https://drive.google.com/drive/folders/1_mIUXzWdfXwy4XapOQedXiA6G9aEutZQ?usp=drive_link`.

Declarations

List of abbreviations

- CNAs: *Copy Number Alterations*
- DL: *Deep Learning*
- ML: *Machine Learning*
- SHAP: *SHapley Additive exPlanations*
- SNNs: *Siamese Neural Networks*
- SNPs: *Single Nucleotide Polymorphisms*
- TCGA: *The Cancer Genome Atlas*
- TMB: *tumor mutational burden*
- TMEs: *tumor microenvironments*

Ethical Approval (optional)

Not applicable.

Consent for publication

Not applicable.

Competing Interests

The author(s) declare that they have no competing interests.

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Author's Contributions

The authors contributed equally to this work.

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Not applicable.

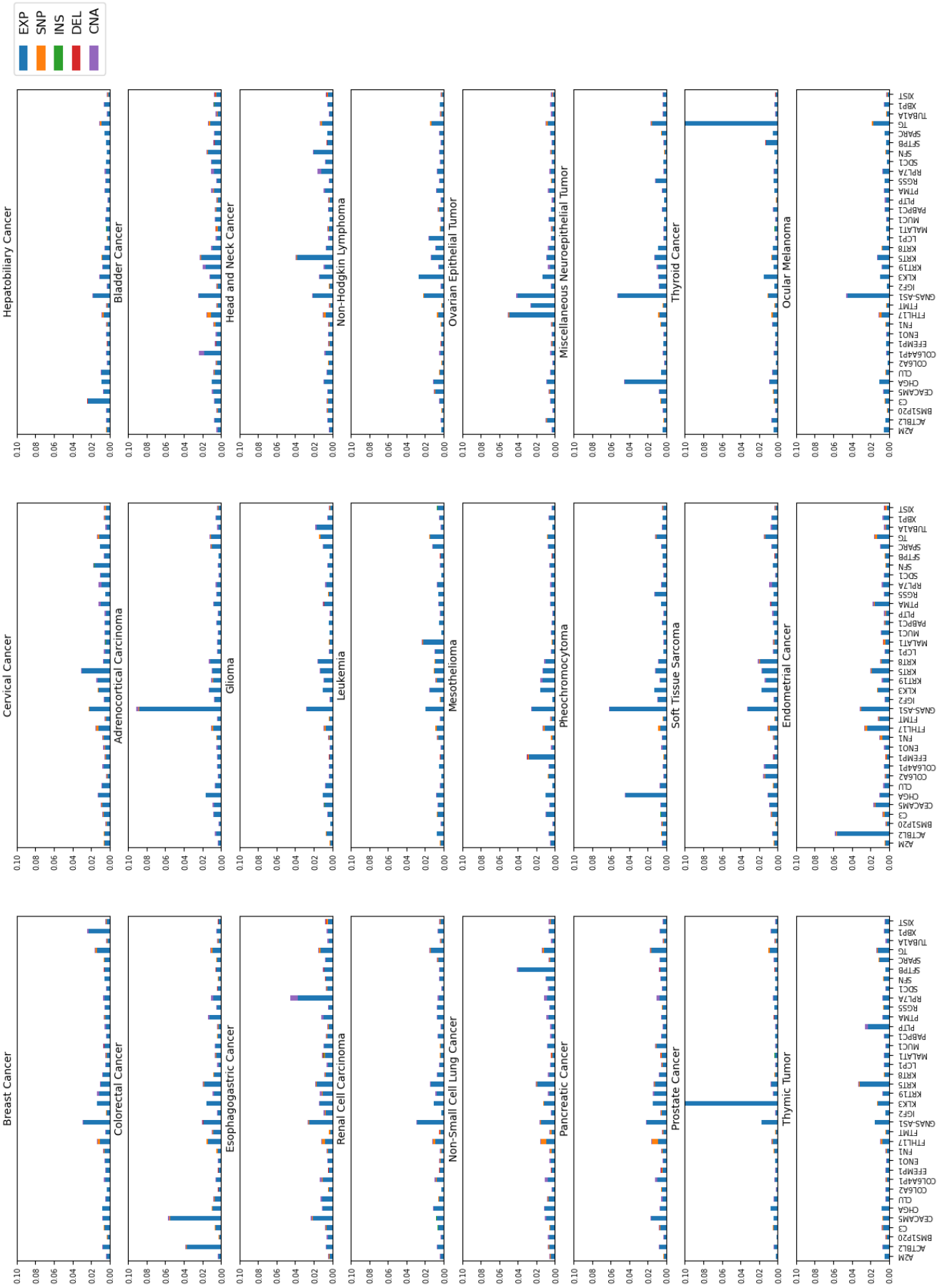


Figure 10. Distribution of *shared genes* importances across different cancer types.

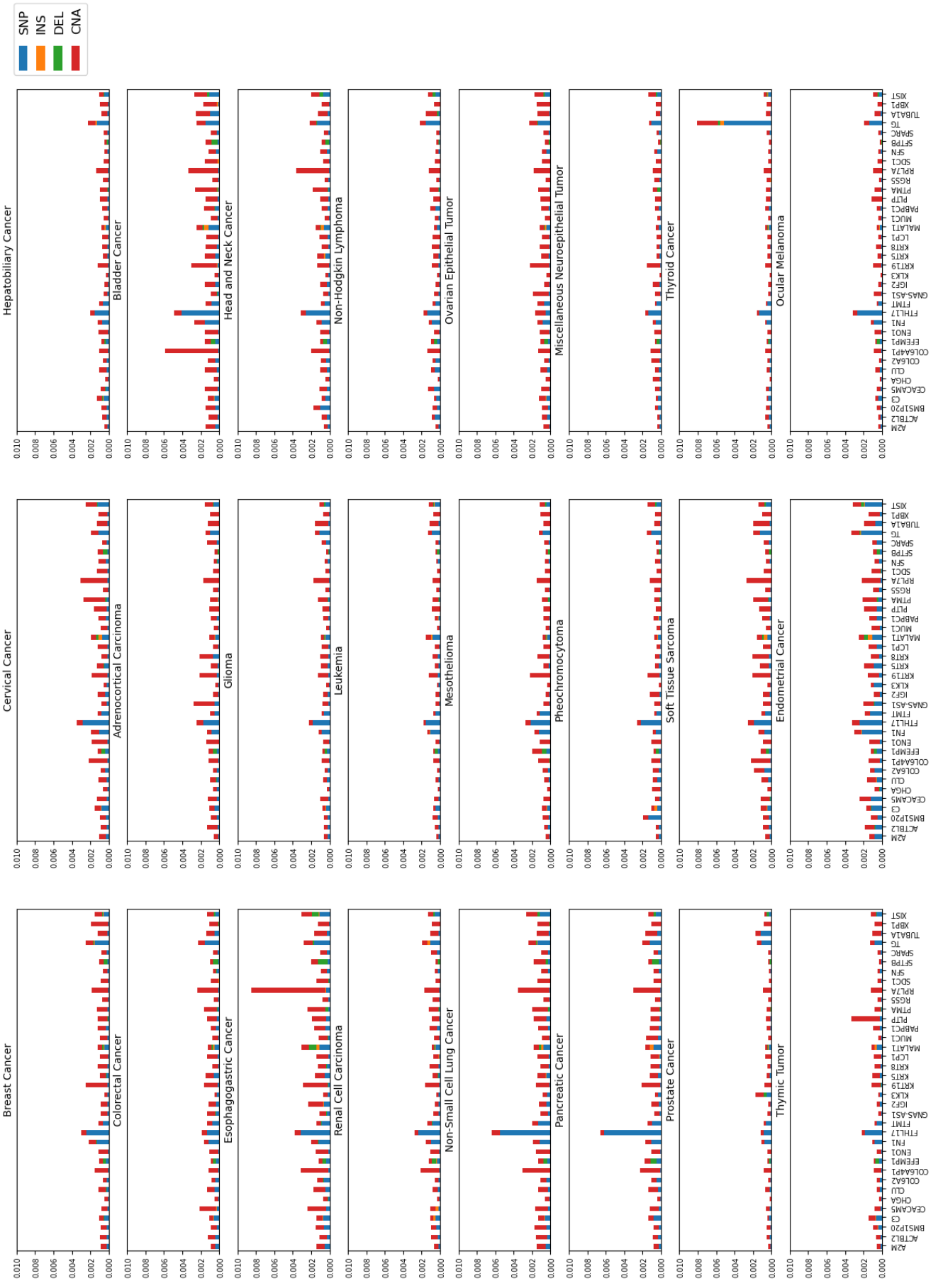


Figure 11. Distribution of *shared genes* **TMB** importances across different cancer types.

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